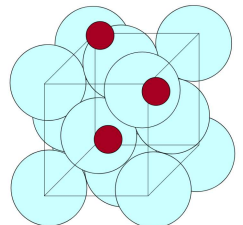


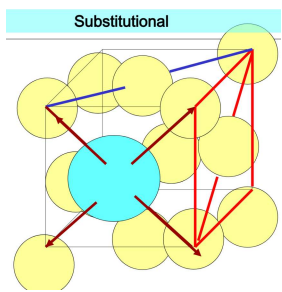
Pure Metals → Ductile / Soft
→ Low lattice resistance

3 Methods to harden metals:

- Solid solution strengthening:
 - Can harden a metal by alloying it (making impure)
 - Can dissolve impurities in a solid crystal - just like in a liquid
 - E.g.:-Copper can dissolve up to 30% Zn
 - Copper and Zn atoms not same size
 - Dissolving one element in solid crystal of another implies:
 - solute fits into spaces between atoms of crystal (Interstitial alloy)



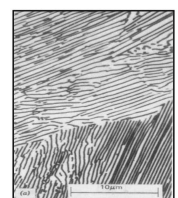
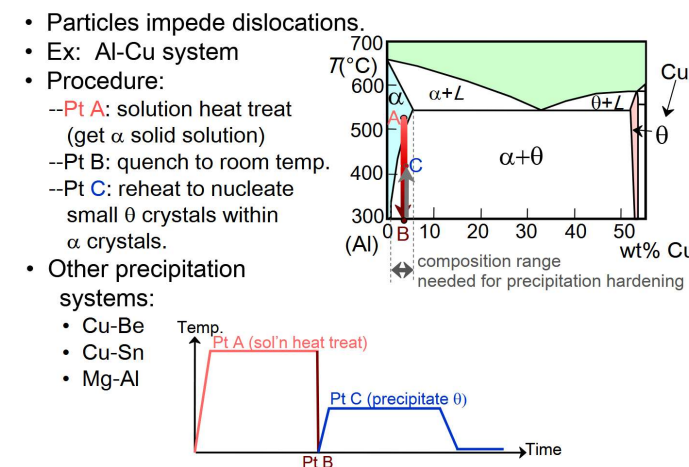
→ Solute replaces atoms of solvent crystal (Substitutional alloy)



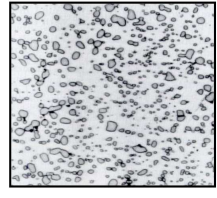
- In each case, the lattice is distorted.
- Introduces stresses and roughens the slip plane
- Makes it harder for dislocations to move.

• Precipitate / dispersion hardening:

- If solution is saturated, change in temp. may result in solute precipitating out of solution.
- Happens in both liquid and solid solutions



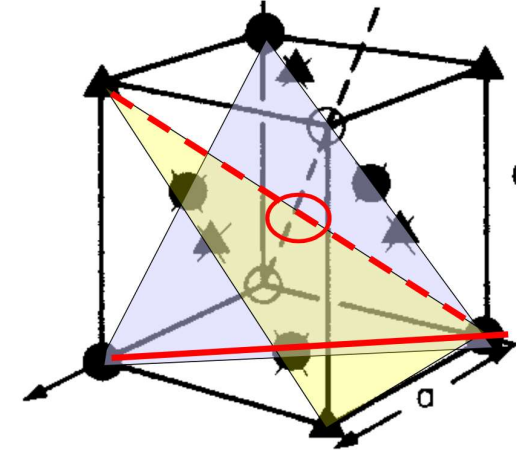
Steel is strengthened by Iron Carbide Precipitates.



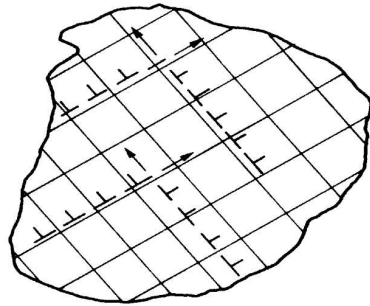
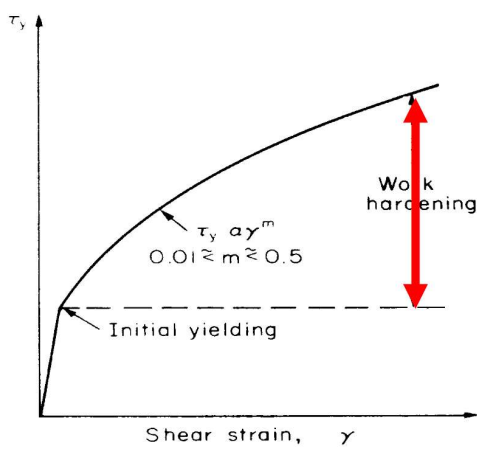
An alloy of aluminium with 4% copper (Duralumin) can be cooled to produce a fine dispersion of fine precipitates of hard CuAl2.

• Work hardening:

- When crystals yield, dislocations move
- Most crystals have several slip planes; fcc crystal has slip planes {1, 1, 1}



Dislocations moving on the different {111} planes will interact, intersect and obstruct each other - raising the stresses required to continue the movement. The dislocations build up and accumulate in the crystal

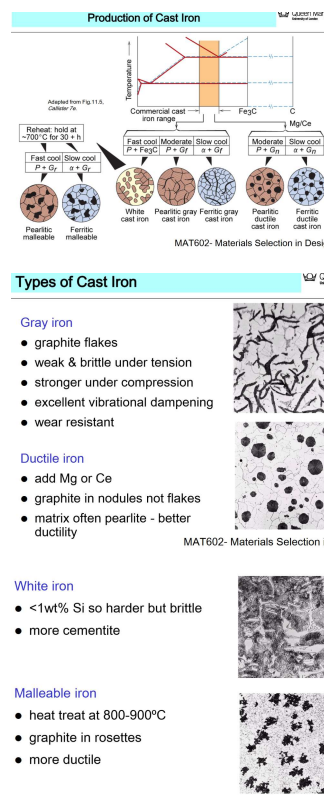
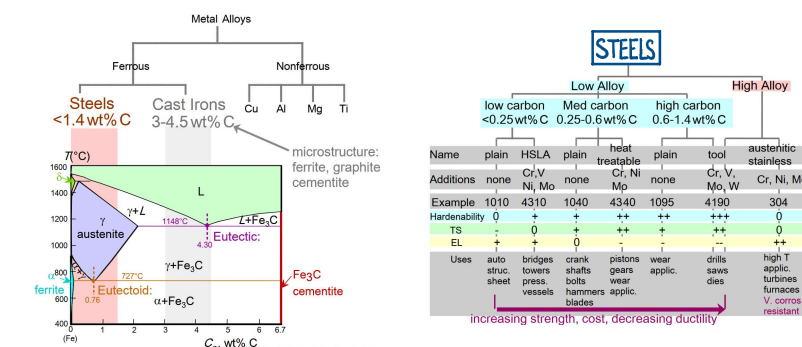


The result is work hardening - the stress required to continue plastic deformation increases after the initial yield.

Metal types:

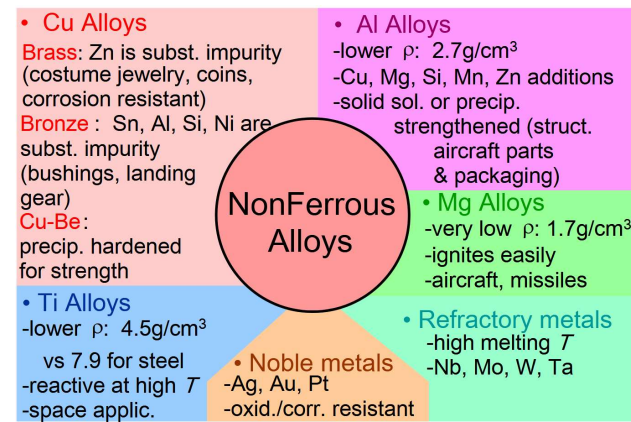
Ferrous alloys

- Iron always a prime constituent
- e.g.
 - low carbon steel, high carbon steel, stainless steel, cast iron, grey iron, white iron.



Non Ferrous Alloys:

- Copper alloys
 - Brasses (Copper & Zinc)
 - Bronges (Copper & tin, aluminium, silicon, nickel)
- Aluminium alloys
 - Aluminium - lithium alloys used in aircraft structures
 - Aluminium - Copper alloys used in chemical + food industries
- Titanium alloys
- Magnesium alloys



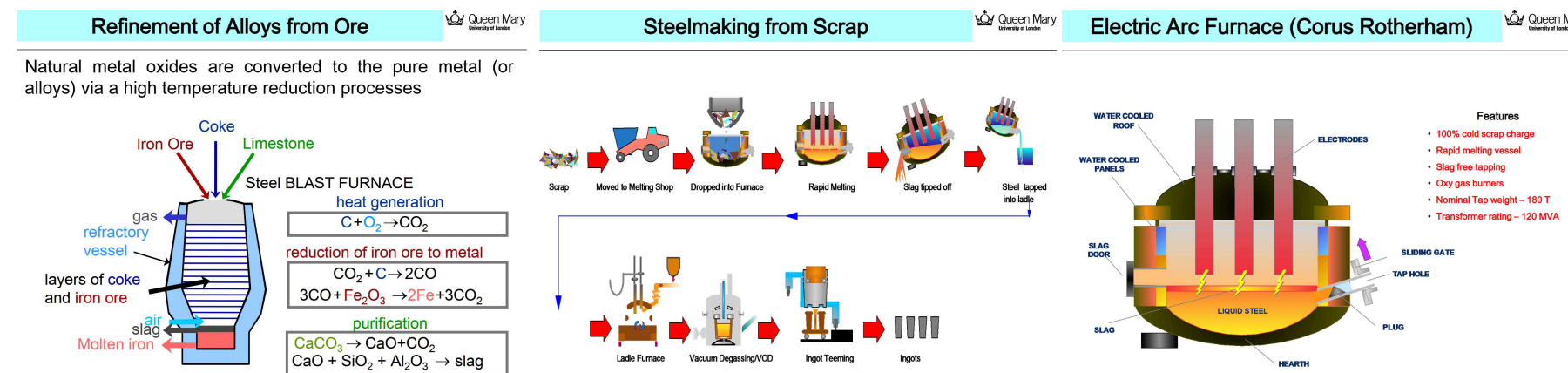
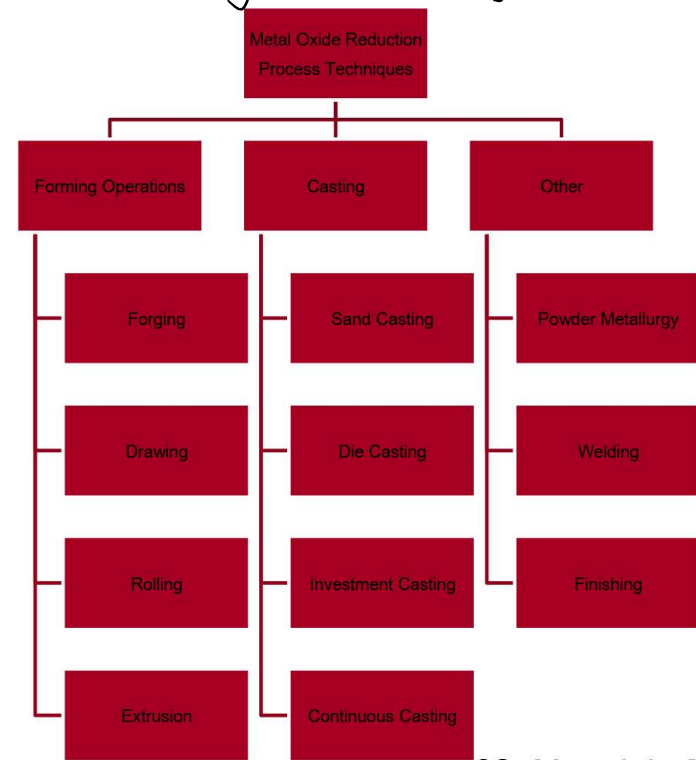
Process Considerations:

- Energy cost taking metal to melt temp. is too high
- Desirable to form or shape during initial transformation from melt
- Shaping operations at melt temps require:
 - Expensive moulds or tools made from exotic metals or materials, difficult to form accurately.

- Cooling from high temps. results in significant shrinkage impacting dimensional tolerances
 - Creates air gap which insulates mould

- Work hardening limits extent of plastic deformation that can occur in a single step.
- Extent of plastic forming is limited
- Metals can't be easily pressed around corners
- Most presses operate in one direction, however liquids flow in all directions
- Phase changes occur on both heating and cooling

Primary alloy processing:



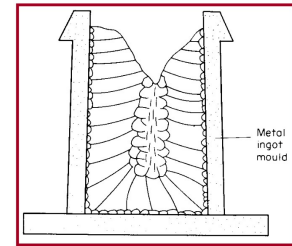
Casting - mould filled with metal

- Metal melted in furnace (alloying elements added), then cast in a mould
- Most common/cheapest method
- Good production of shapes
- Weaker products / internal defects
- Good option for brittle materials

Ingot Casting:

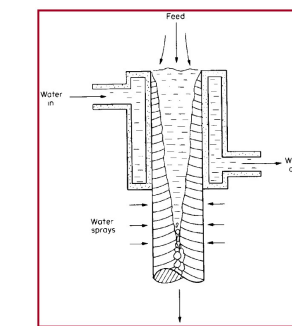
- Processes depend on material produced
- Mould type depends on material, hence melt temp.
- Moulds usually simple with rounded corners to allow ease of removal.
- Metals have a grain structure
- Shrinkage → Contraction cavity at top of ingot.
- Impurities first expelled from solid and concentrated in last melt zones
- Gas bubbles may form from dissolved nitrogen released on cooling

Metal	Melt temp/°C
Tungsten	3400
Silica	1900
Platinum	1770
Iron	1536
Nickel	1453
Mild steel	1413
Gold	1063
Copper	1083
Aluminium	660
Magnesium	650
Zinc	420
Lead	327



Continuous Casting:

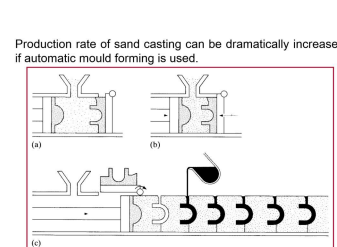
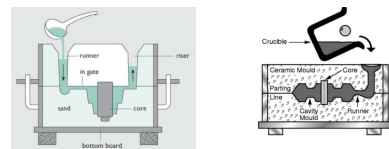
- Many problems eliminated if material is converted into shape directly during continuous casting process to make bar or sheet
- Contraction cavities don't form as material is continuously replenished.
- Segregation reduced because columnar grains grow over smaller distances.
- Work required to convert product into rod or sheet is less because starting block is smaller



3 Approaches:

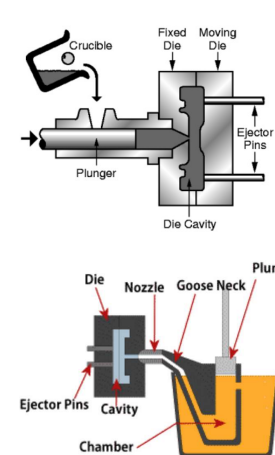
Permanent patterns (Sand casting)

- Pattern / model of part used to make impression in expendable mould
- Mould destroyed upon each use, pattern reused

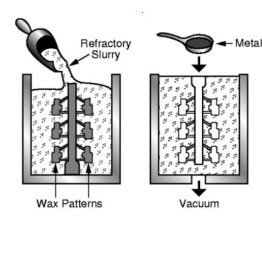
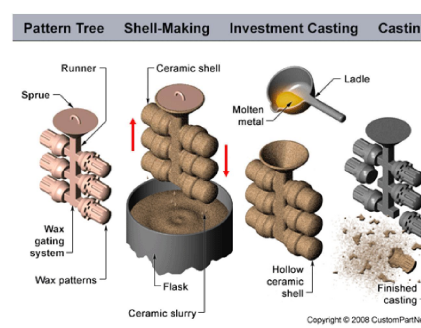


Permanent patterns (Die casting)

- Mould reused; part released by opening mould rather than destroying it
- Moulds made accurately from materials that can withstand high temps.
- Restricts process to low melt temp alloys
- Moulds are expensive
- To be used economically, requires:
 - Long production runs
 - Or, small components



Expendable mould and pattern (Investment casting):



Forming:

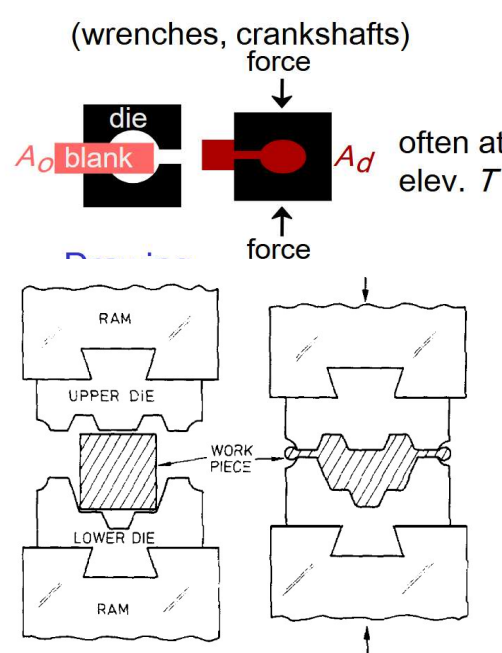
- Shape changing operations; Convert shape via plastic deformation
- Why use forming at all if we have casting available?
 - There are **defects in castings** that for a cost can be reduced.
 - Geometry - Long (one dimension significantly greater than the others) shapes such as tubes are difficult to cast
 - Microstructure - Best castings come from eutectic alloys and this is not always desirable.
 - Inhomogeneity - common in castings but plastic deformation (the basis for forming) reduces inhomogeneity.
 - Refractory materials - Some materials are difficult to process as liquid due to high melt temperatures - usually processed as powders with some associated forming operations.

Classification:

- Continuous = Rolling
 - Bulk deformation
 - Trivial stresses
- Intermittent = Forging
 - Starting material → high v. to s.s.

Forging (Hammering, stamping):

- Hot forging → bigger shape change, but poor surface & tolerance because of oxidation & warpage.
- Cold forging → greater precision & finish, but forging pressures are higher and deformations limited by work hardening.



- Formability = min radius to which a sheet can be bent.
 - Sometimes expressed in terms of sheet thickness t (1 = good; 4 = average)
 - Determines amount sheet can be stretched or drawn without necking and failing

- Radius best made as large as possible; never less than sheet thickness
- Forming limit diagrams give more precise info - shows combination of principle strains in plane of sheet that cause failure
 - Parts designed so strains don't exceed this limit

Rolling (Hot and Cold):

Cold rolling: ($T < 0.35 T_m$)

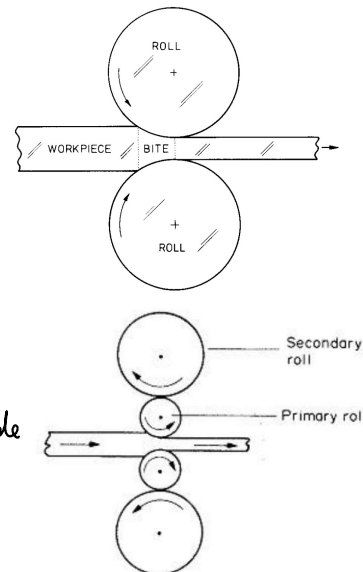
- Exploits work hardening to increase strength of final product; at penalty of higher forming pressures.

Hot rolling

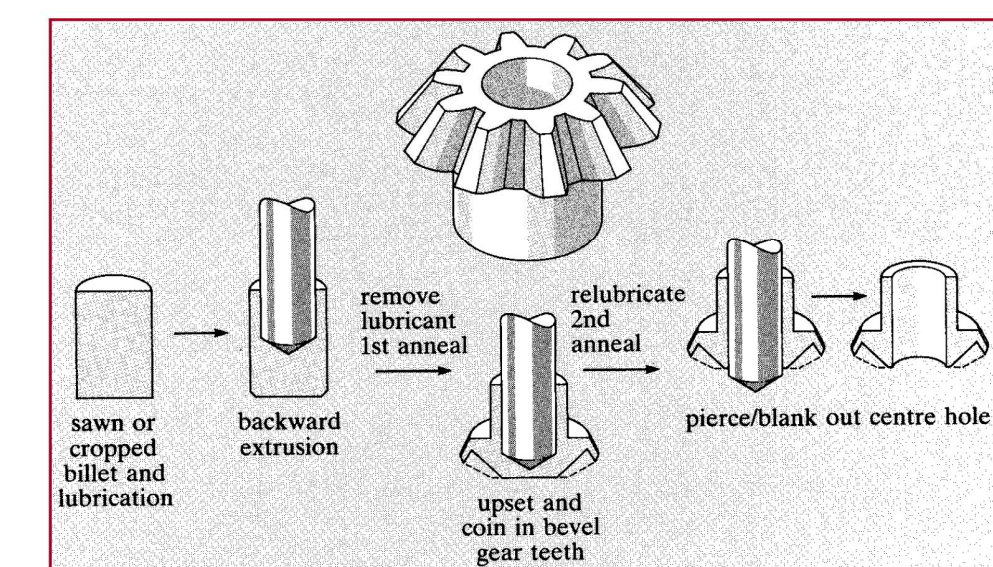
- Lower work hardening + lower pressures

Roller geometry:

- Primary roller should always be as small as possible
- Secondary roller (if present) can be bigger



FORGING OF A GEAR



Material Considerations

- Real metals exhibit work hardening; limits extent of deformation possible, also increases hardness of material.
- Basis for difference between cold and hot working
 - Slipp; hard & strong
- However, forces needed to deform metal are high; resulting in in-built stresses and corrosion during service

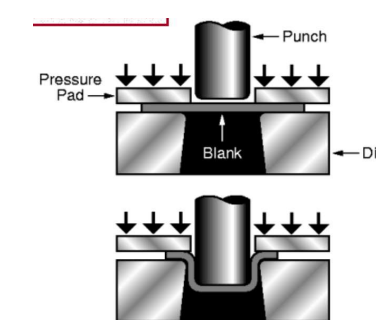
TRI-AXIAL COMPRESSION

- Q. Why do forming operations usually involve tri-axial compression?

- Hydrostatic compression (tri-axial with balanced $\sigma_1, \sigma_2, \sigma_3$) involves no shear but can result in overall volume change.
- Mismatch in compressive stresses introduces shear stresses on some planes - these result in plastic deformation.
- Key point is that the hydrostatic component of the pressure ensures that defects do not propagate as tensile stresses would lead to crack growth, rupture or tearing.

Drawing: (Usually only biaxial forces)

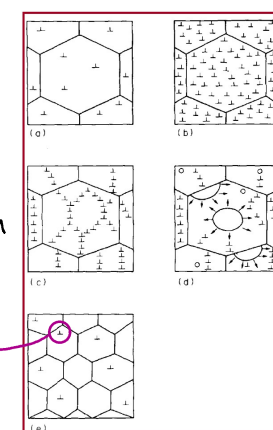
- Blanking - cutting via shear forces punch and die
- Deep drawing - bending, reverse bending
- No change in volume - side walls may increase in thickness
- Deformation limited due to side walls forces which will ultimately fracture



Hot working:

- Can exhibit:
 - reduced yield stress
 - recovery and re-crystallisation

Dislocation

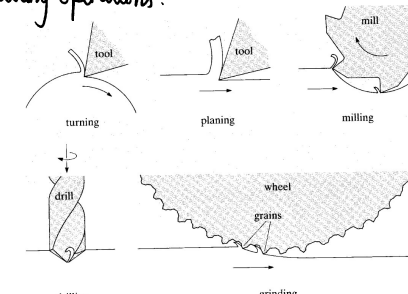


Machining:

- Secondary process, used if:
 - Introduction of holes, slots, etc; difficult via primary process
 - To improve dimensional tolerances or surface texture
- Sometimes a primary process (CNC, milling, grinding)
- Tooling costs reduced; if material is cheap, then wastage tolerated

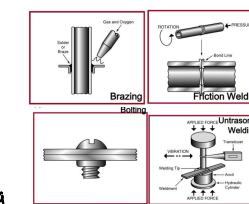
Cutting:

- Cutting tool harder than work material, and work piece ductile
- Single point cutting operations:
 - turning
 - planing
 - milling
 - drilling
 - grinding



Joining:

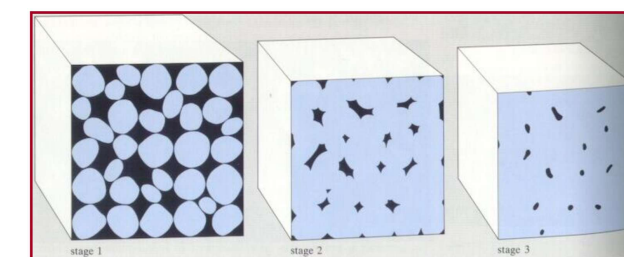
- Welding
- Fastening
- Adhesive bonding
- Bolts and screws (facilitate disassembly)



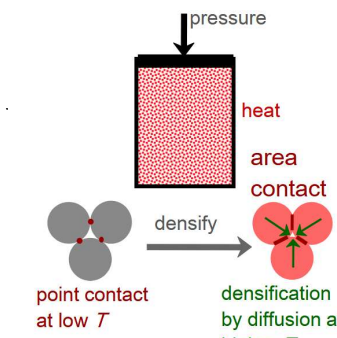
Powder metallurgy:

- Liquid metal changed to powder via atomisation process.

- Can be compacted into shapes & sintered
- Used with high melt temp materials
- Processes near net shape, reducing finishing operations
- Good if hard materials like carbides & nickel super-alloys being processed
- Unusual combos of metals/alloys possible
- Unusual / Unstable microstructures (rapidly solidified alloy products)



Compacting / Pressure forcing particles together



Thermal Processing of Metals

Annealing: Heat to T_{anneal} , then cool slowly.

- Stress Relief: Reduce stress caused by:
 - plastic deformation
 - nonuniform cooling
 - phase transform.
- Spheroidize (steels): Make very soft steels for good machining. Heat just below T_F & hold for 15-25h.
- Full Anneal (steels): Make soft steels for good forming by heating to get γ , then cool in furnace to get coarse P .
- Process Anneal: Negate effect of cold working by (recovery) recrystallization)
- Normalize (steels): Deform steel with large grains, then normalize to make grains small.

Types of Annealing

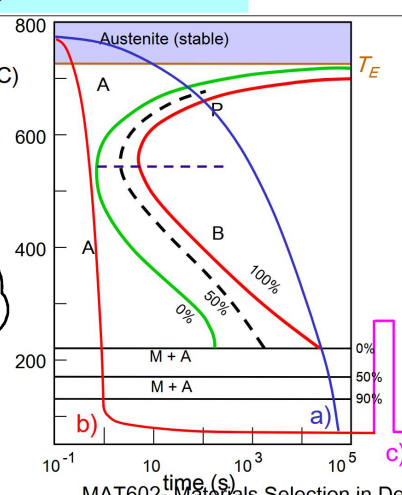
Heat Treatments

a) Annealing

b) Quenching

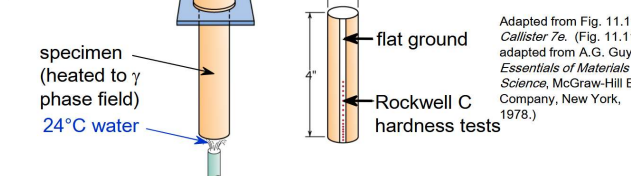
c) Tempered Martensite

TTT Graph

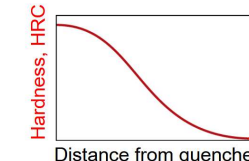


Hardenability-Steels

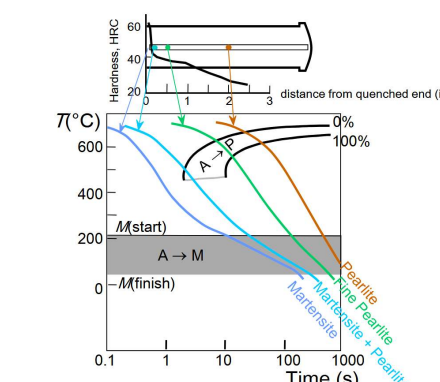
- Ability to form martensite
- Jominy end quench test to measure hardenability.



• Hardness versus distance from the quenched end.

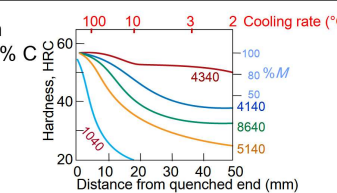


- The cooling rate varies with position.



Hardenability versus Alloy Composition

- Jominy end quench results, $C = 0.4 \text{ wt\%}$



"Alloy Steels"

- (4140, 4340, 5140, 8640)
- contain Ni, Cr, Mo (0.2 to 2wt%)
- these elements shift the "nose"
- martensite is easier to form.

